

T.POOJITHA (192412318)

ITA0609

Exp-6: Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

```
# Import required libraries

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import confusion_matrix, accuracy_score

# Load the Iris dataset

iris = load_iris()

X = iris.data    # Features

y = iris.target  # Target labels

# Split the dataset into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create the Naïve Bayes model

model = GaussianNB()

# Train the model

model.fit(X_train, y_train)

# Predict the test data

y_pred = model.predict(X_test)

# Display Actual and Predicted values

print("Actual Labels :", y_test)

print("Predicted Labels:", y_pred)

# Generate Confusion Matrix

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")

print(cm)
```

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# Calculate Accuracy
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accuracy = accuracy_score(y_test, y_pred)
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print("\nAccuracy: {:.2f}%".format(accuracy * 100))
```

Output:

```
... Actual Labels : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Predicted Labels: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

Accuracy: 100.00%
```